

Cover Letter

September 08, 2026

Dear Editors,

Please consider our manuscript entitled "Provenance-Controlled Graph Context Augmentation for Auditable Drug-Adverse Effect Relation Extraction" for publication in the Journal of Safety Science and Resilience.

The study addresses two connected questions in graph-augmented extraction: which retrieved associations are eligible to guide generation, and which generated candidates should enter the final output. A provenance-constrained graph context policy supplements source text and the label ontology with concept hints, source-matched anchors, relation priors, and retrieved semantic relation patterns. It uses training annotations only, excludes current-example-only support, and quarantines exact cross-split text overlaps. A separate evidence-gated acceptance mechanism combines source-span reconstruction, entity selection, relation verification, and endpoint consistency with inspectable decisions.

Under a common strict-span evaluation, the proposed system improves relation F1 over matched source-only generation by 5.18 percentage points on ADE and 4.65 points on CoNLL04. The relation gains recur in two additional training runs and persist after exact CoNLL04 train-test overlaps are excluded. Development-set ablations distinguish candidate-recovery benefits from the precision and coverage effects of deterministic acceptance.

The primary application is drug-adverse-effect extraction from medical text, evaluated on ADE. CoNLL04 provides a supplementary general-domain benchmark to examine applicability under different entity roles and relation structures. The datasets are trained separately; this is not a direct cross-domain transfer experiment. The connection to safety science is inspectable knowledge construction for drug-safety-related information. The framework exposes source links and acceptance decisions for analyst review. The study evaluates extraction and rule compliance, not clinical truth or measured downstream safety outcomes.

Thank you for considering our manuscript.

Sincerely,

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